
Probability and Statistic (for CS Students)

Lecture 3: Counting and Combinatorics II

University of Mazandaran

Fall 1405

Main rules of counting:

▶ **Step rule of counting**

- ▶ the experiment runs in r steps
- ▶ the first step has n_1 possible outcomes, and
- ▶ the second step has n_2 possible outcome, and
- ▶ ...
- ▶ the r^{th} step has n_r possible outcome,
- ▶ number of outcomes in the k^{th} step is unaffected by outcomes of the its previous steps,
- ▶ the number of outcomes of the experiment $n_1 \times n_2 \times \cdots \times n_r$

▶ **Or rule of counting** (for two outcome sets)

- ▶ the outcome of an experiment can either be drawn from set A or set B
- ▶ the sets may potentially overlap
- ▶ the number of outcomes of the experiment $|A| + |B| - |A \cap B|$

Question: A 6-bit string is sent over a network. The valid set of strings recognized by the receiver must either start with “01” or end with “10”. How many such strings are there?

Answer:

$$\begin{aligned}
 N &= |A| + |B| - |A \cap B| \\
 &= 16 + 16 - 4 \\
 &= 28
 \end{aligned}$$

2^4 start with 01

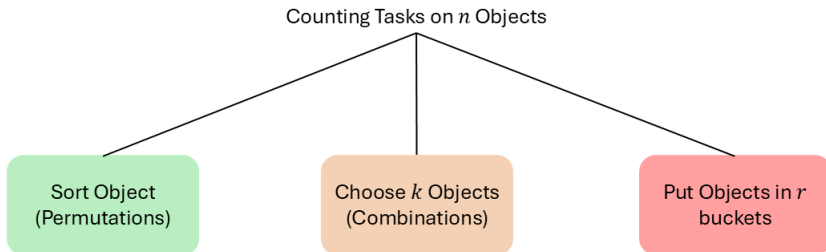
010000
010001
010010
010011
010100
010101
010110
010111
011000
011001
011010
011011
011100
011101
011110
011111

Set A

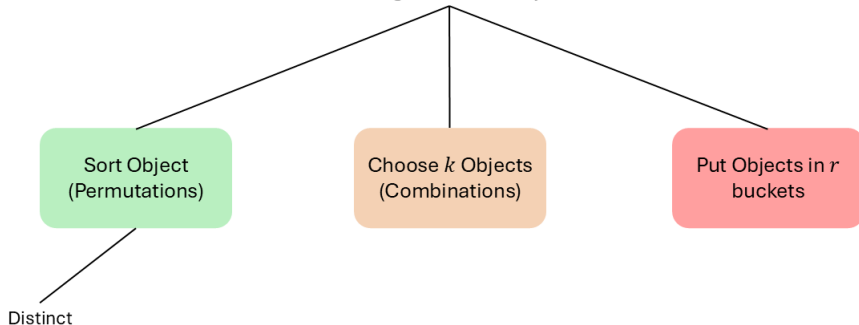
2^4 end with 10

000010
000110
001010
001110
010010
010110
011010
011110
100010
100110
101010
101110
110010
110110
111010
111110

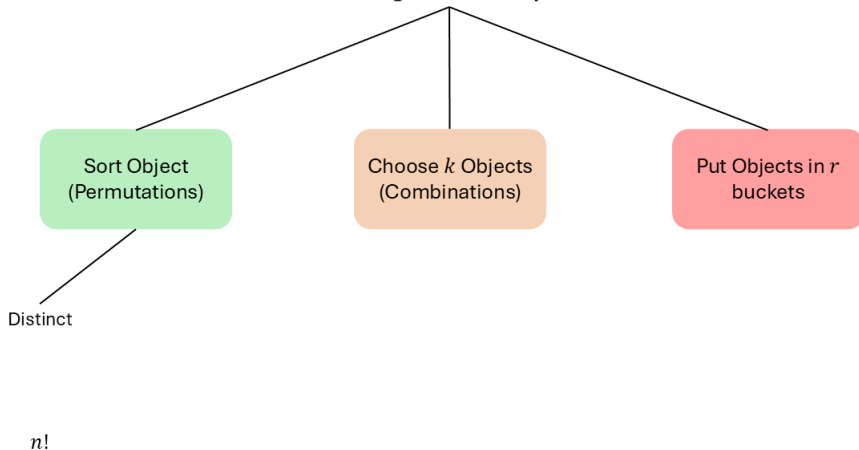
Set B



Counting Tasks on n Objects

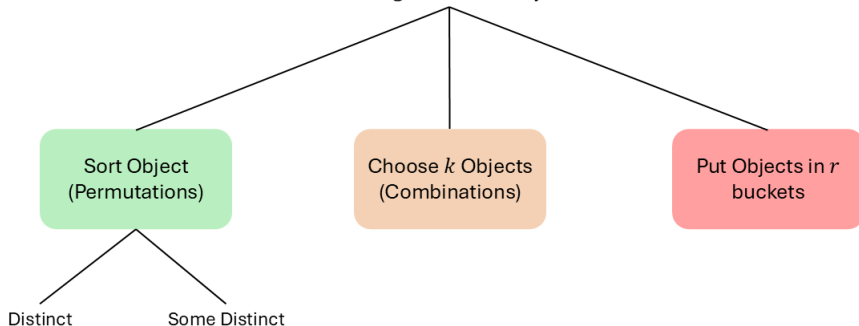


Counting Tasks on n Objects



Permutation With Indistinct Objects

Counting Tasks on n Objects



$n!$

How Many Unique Bit Strings?

1,0,1,0,0

How Many Unique Bit Strings?

Let's simplify!

1, 0, 1, 0, 0

How many unique bit strings do we have now?

How Many Unique Bit Strings?

Let's simplify!

1, 0, 1, 0, 0

How many unique bit strings do we have now? $5! = 120$

How Many Unique Bit Strings?

So, if bits were distinct, we could generate $5!$ unique strings.
BUT THEY ARE NOT!

1,0,1,0,0

How many repeated strings do we have?

How Many Unique Bit Strings?

So, if bits were distinct, we could generate $5!$ unique strings.
BUT THEY ARE NOT!

1,0,1,0,0

How many repeated strings do we have? $2! \times 3!$

Permutation With Indistinct Objects

How do we find the number of permutations considering some objects are indistinct

$$\text{permutations considering some objects are indistinct} = \frac{\text{permutations of distinct objects}}{\text{permutations of just the indistinct objects}}$$

General Rule to Counting Permutations

When there are n objects such that

- ▶ n_1 are the same (indistinguishable or **indistinct**), and
- ▶ n_2 are the same, and
- ...
- ▶ n_r are the same,

the number of unique orderings (**permutations**) is

$$\frac{n!}{n_1! n_2! \cdots n_r!}$$

An international tournament has 10 competitors, of which 4 are Iranian, 3 are Turkish, 2 are Afghan, and 1 from Iraq. If the tournament result lists just the nationalities of the palyers in the order in which they placed, how many outcomes are possible?

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$$\frac{10!}{4!3!2!1!} = 12600$$

Python code

```
import itertools

def main():
    letters=['A', 'A', 'N', 'N', 'N', 'N', 'Q', 'T', 'T', 'T']
    perms=set(itertools.permutations(letters))

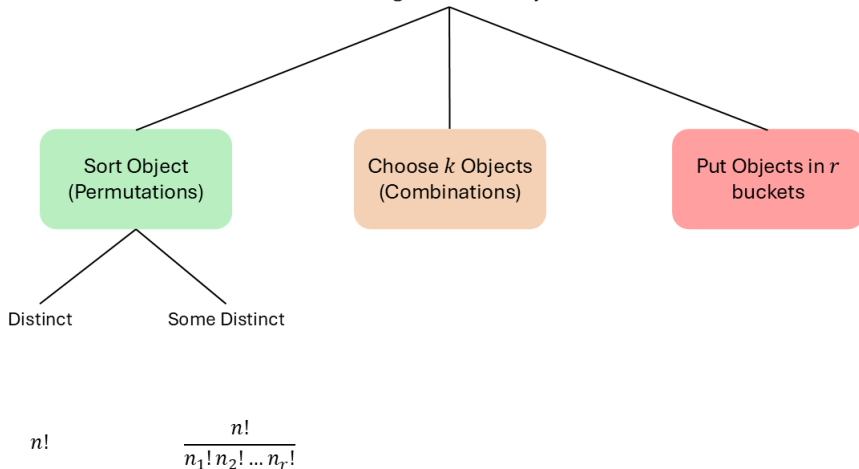
    for perm in perms:
        desired_perm="".join(perm)
        print(desired_perm)

import math

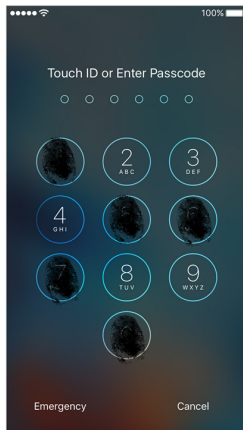
def main():
    n=math.factorial(10)
    d=math.factorial(2)*math.factorial(4)*math.factorial(3)
    print(n/d)
```

Summary of Combinatorics

Counting Tasks on n Objects

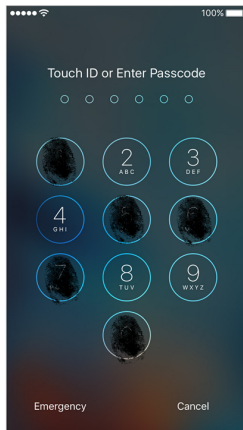


Passcodes



How many unique 6-digit passcodes are possible if a phone password uses 5 distinct numbers?

Passcodes

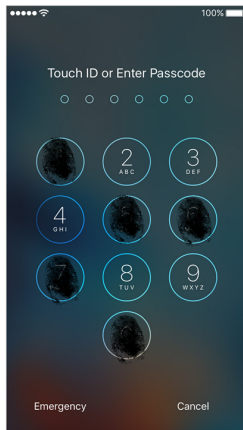


How many unique 6-digit passcodes are possible if a phone password uses 5 distinct numbers?

Steps:

1. choose digit to repeat 5 outcomes
2. create passcode sort 6 digits:
4 distinct, 2 indistinct

Passcodes



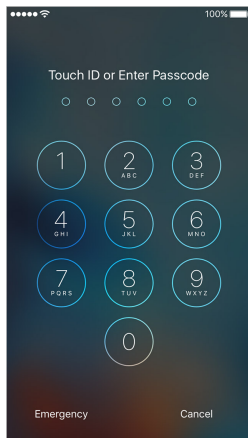
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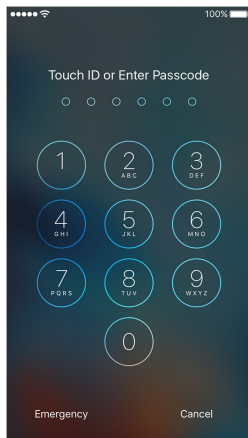
$$\text{Total} = 5 \times \frac{6!}{2!}$$

Passcodes



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Passcodes

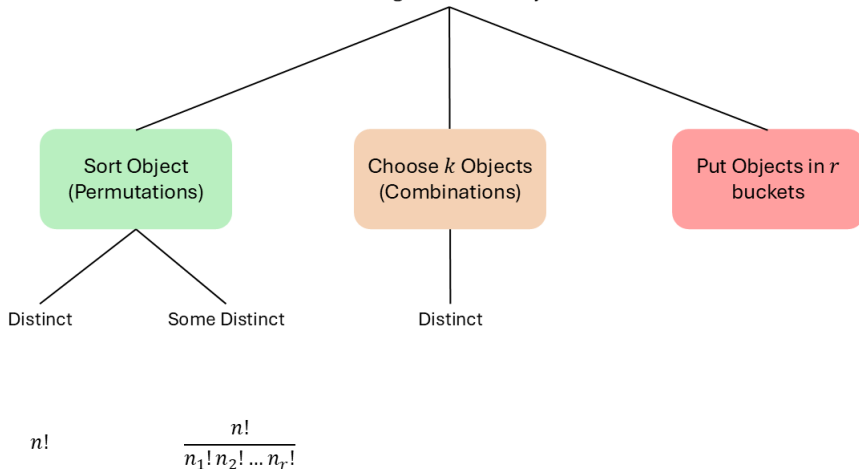


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Steps:

1. choose 5 distinct digits ? outcomes
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Counting Tasks on n Objects



Combinations with Cake

There are $n = 20$ people.

How many ways can we choose $k = 5$ people to get cake?



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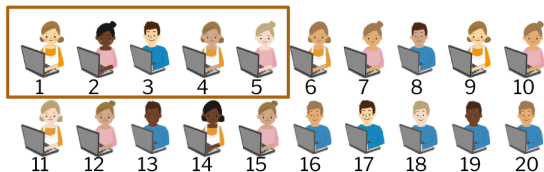
1. n people get in line

$n!$ ways

Combinations with Cake

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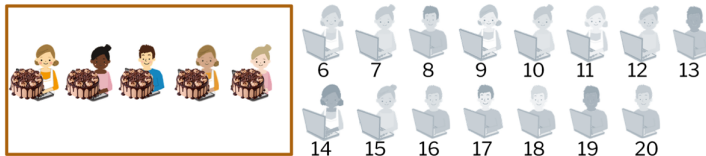
► Step 1: choose 5 out of 20

1. n people get in line $n!$ ways
2. put the first k in cake room 1 way

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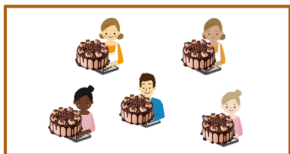
How many ways can we choose $k = 5$ people to get cake?

- ▶ Step 1: choose 5 out of 20
 - 1. n people get in line $n!$ ways
 - 2. put the first k in cake room 1 way
- ▶ Step 2: remove repetitions

Combinations with Cake

There are $n = 20$ people.

How many ways can we choose $k = 5$ people to get cake?



► Step 1: choose 5 out of 20

1. n people get in line

$n!$ ways

2. put the first k in cake room

1 way

► Step 2: remove repetitions

3. allow cake group to mingle

$k!$ ways

Combinations with Cake

There are $n = 20$ people.

How many ways can we choose $k = 5$ people to get cake?



► Step 1: choose 5 out of 20

1. n people get in line $n!$ ways
2. put the first k in cake room 1 way

► Step 2: remove repetitions

3. allow cake group to mingle $k!$ ways
4. allow non-cake group to mingle $(n - k)!$ ways

Combinations

A **combination** is an unordered selection of k objects from a set of n **distinct** objects.

The number of ways of making this selection is:

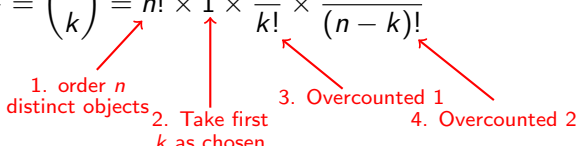
$$\frac{n!}{k!(n-k)!} = \binom{n}{k} = n! \times 1 \times \frac{1}{k!} \times \frac{1}{(n-k)!}$$

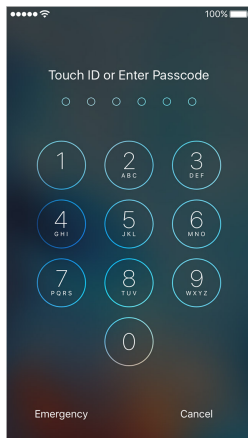
1. order n distinct objects

2. Take first k as chosen

3. Overcounted 1

4. Overcounted 2

The diagram illustrates the derivation of the combination formula. It shows the equation $\frac{n!}{k!(n-k)!} = \binom{n}{k} = n! \times 1 \times \frac{1}{k!} \times \frac{1}{(n-k)!}$. Four red arrows point from explanatory text to specific parts of the formula: 1. An arrow points from '1. order n distinct objects' to the $n!$ term. 2. An arrow points from '2. Take first k as chosen' to the '1' term. 3. An arrow points from '3. Overcounted 1' to the $\frac{1}{k!}$ term. 4. An arrow points from '4. Overcounted 2' to the $\frac{1}{(n-k)!}$ term.



How many unique 6-digit passcodes are possible if a phone password uses 5 distinct numbers?

Steps:

1. choose 5 distinct digits $\binom{10}{5}$ outcomes
2. choose digit to repeat 5 outcomes
3. create passcode sort 6 digits:
4 distinct, 2 indistinct

$$\text{Total} = \binom{10}{5} \times 5 \times \frac{6!}{2!}$$

Committee Selection

From a group of 5 women and 7 men, how many different committees consisting of 2 women and 3 men can be formed if 2 of the men refuse to serve on the committee together?

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From a group of 5 women and 7 men, how many different committees consisting of 2 women and 3 men can be formed if 2 of the men refuse to serve on the committee together?

Approach 1: direct solution

From the 2 men refuse to serve together, either one of them could be in the committee or non of them

► One of them be in the committee

step 1: choose one of the two men $\binom{2}{1}$

step 2: choose the other two from the remaining 5: $\binom{5}{2}$

step 3: choose two women from the team of five: $\binom{5}{2}$

► None of them be in the committee

step 2: choose all 3 men from the remaining 5: $\binom{5}{3}$

step 3: choose two women from the team of five: $\binom{5}{2}$

Total number of committees: $\binom{2}{1} \binom{5}{2} \binom{5}{2} + \binom{5}{3} \binom{5}{2}$

From a group of 5 women and 7 men, how many different committees consisting of 2 women and 3 men can be formed if 2 of the men refuse to serve on the committee together?

Approach 2: indirect solution

Total possible committees without any constraint: $\binom{5}{2} \binom{7}{3}$

Number of impossible committees (committees with the two men):

step 1: the two men should be in the committee 1

step 2: choose the third man from the remaining 5: $\binom{5}{1}$

step 3: choose two women from the team of five: $\binom{5}{2}$

Total number of committees: $\binom{5}{2} \binom{7}{3} - \binom{5}{1} \binom{5}{2}$

How many ways are there to choose 3 books from a set of 6 distinct book?

How many ways are there to choose 3 books from a set of 6 distinct book?

$$\binom{6}{3} = \frac{6!}{3!3!} = 20 \text{ ways}$$

Python code

```
import itertools

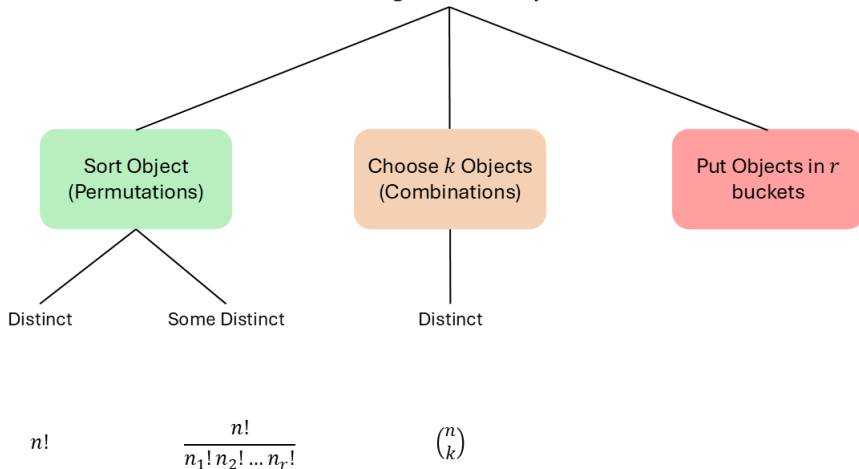
def main():
    books = ['b1', 'b2', 'b3', 'b4', 'b5', 'b6']
    chosen=itertools.combinations(books,3)
    for book in chosen:
        print(book)

import math

def main():
    total = math.comb(6, 3)
    print(total)
```

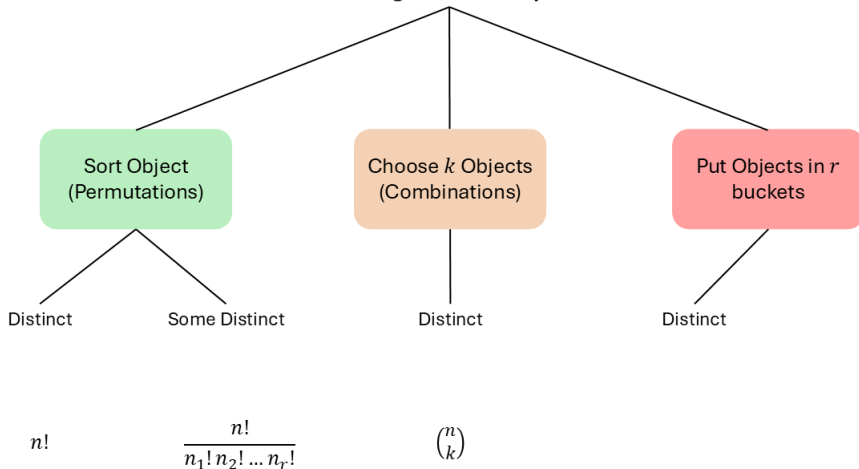
Summary of Combinatorics

Counting Tasks on n Objects



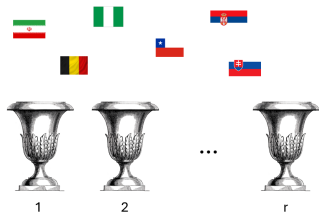
Summary of Combinatorics

Counting Tasks on n Objects



World Cup Seeding and Combinatorics!

How many ways are there to hash n **distinct** objects to r buckets?

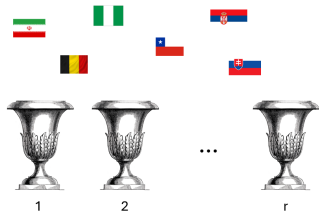


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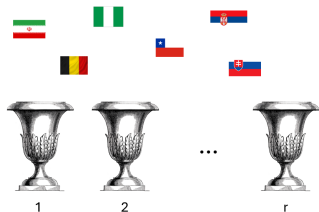
Steps:

1. Buckets for the 1th flag $\rightarrow r$ choices
2. Buckets for the 2nd flag $\rightarrow r$ choices
- ...
- n . Buckets for the n^{th} flag $\rightarrow r$ choices



World Cup Seeding and Combinatorics!

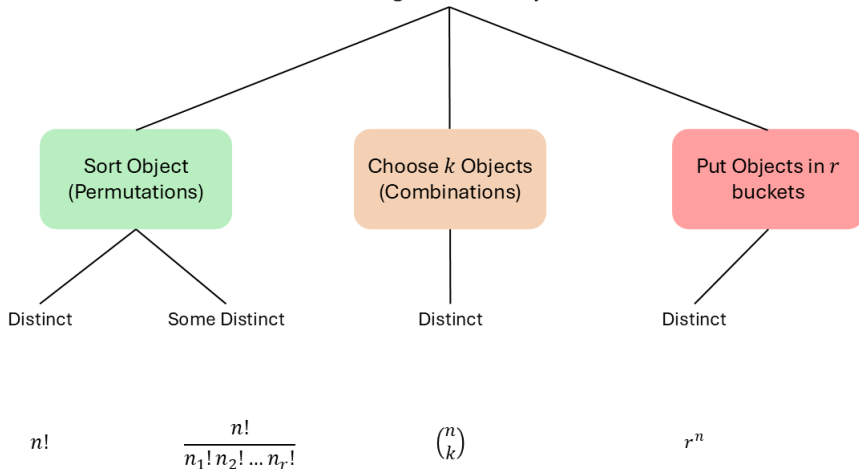
How many ways are there to hash n **distinct** objects to r buckets?



r^n possibilities

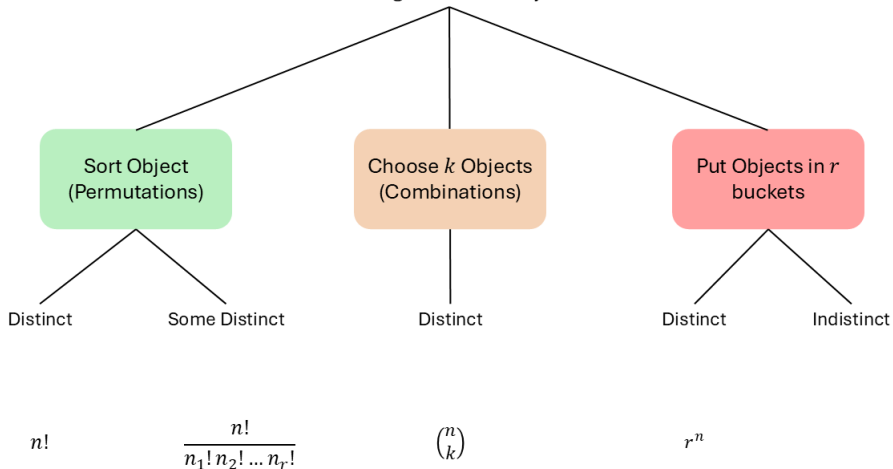
Summary of Combinatorics

Counting Tasks on n Objects



Summary of Combinatorics

Counting Tasks on n Objects



Indistinct Case

You are rich! You want to invest a small portion of your money, say 10M \$, in 4 companies by dividing the investment into 1M \$ portions. How many ways are there to allocate this money?

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Answer: $10\text{M \$} \equiv 10 \times 1\text{M\$}$

Question: 10 indistinct object, 4 positions

Dividers: using 3 indistinct dividers, we can create 4 positions

Question reformed: 10+3 objects including 10 indistinct object of type A and 3 indistinct objects of type B

Solution:

$$\frac{(10 + 4 - 1)!}{10!3!}$$

The number of ways to distribute n indistinct objects into r buckets is equivalent to the number of ways to permute $n + r - 1$ objects such that

- ▶ n are indistinct objects, and
- ▶ $r - 1$ are indistinct dividers

$$\text{Total} = (n + r - 1)! \times \frac{1}{n!} \times \frac{1}{(r - 1)!} =$$
$$\binom{n + r - 1}{r - 1}$$

Summary of Combinatorics

Counting Tasks on n Objects

